One-Way ANOVA — Part A

Introduction to One-Way ANOVA Instructor Edition

Name: _____ **Date:** April — (Instructor Key)

Scenario: A professor in the College of Science and Mathematics at Cal Poly SLO wants to understand whether the **study method** a student uses affects their performance on a unit exam in introductory statistics. She randomly assigns 30 students to one of three study conditions: **(A)** traditional re-reading of notes, **(B)** active recall with flashcards, **(C)** AI-assisted study with structured prompting. After the unit, all students take the same 100-point exam. The professor records each score and wants to determine whether differences in mean performance across groups could be explained by random variation alone, or whether there is real evidence of an effect.

1 Research Question

State the research question for this study in a complete sentence. Your question should identify what is being compared and for whom.

Question *Write the research question*

✓ Is there a difference in mean exam score across the three study methods
✓ (re-reading, active recall, AI-assisted) among students in introductory
✓ statistics at Cal Poly SLO?

2 Variables

(a) State the variables

List each variable mentioned or implied in the scenario.

Variable Name	Description
Study Method	The assigned study condition: re-reading (A), active recall (B), or AI-assisted (C)
Exam Score	Score on the unit exam, measured 0–100 points

(b) Identify the Explanatory and Response Variables

Role	Variable (name it)
Explanatory (EV)	Study method (A, B, or C)
Response (RV)	Exam score (0–100 points)

(c) Type of Each Variable

Variable	Type	How many categories / units?
Study method (A, B, or C)	Categorical	3 levels: A, B, C
Exam score (0–100 points)	Quantitative	Points (0–100)

3 Hypotheses

State the null and alternative hypotheses using proper notation. Let μ_A, μ_B, μ_C denote the population mean exam score for each study method group.

Null Hypothesis (H_0)

✓ H_0: μ_A = μ_B = μ_C
✓ The population mean exam score is the same across all three study method groups.

Alternative Hypothesis (H_a)

✓ H_a: At least one μ_i is different from the others.
✓ At least one study method is associated with a different population mean exam score.

> **Caution:** The alternative hypothesis for ANOVA does not specify **which** group differs — only that **at least one** mean is different. A significant result requires follow-up (post-hoc) tests to identify which pairs differ.

4 Analysis

The ANOVA output table below summarizes results from this study. Use it to answer the questions that follow.

Source	df	SS	MS	F	p-value
Between Groups	2	428.5	214.3	9.14	0.003
Within Groups (Error)	27	633.8	23.5		
Total	29	1062.3			

(a) Identify the test statistic

The test statistic for a one-way ANOVA is the **F-statistic**. From the table:
F = **9.14** This means between-group variability is approximately **9 times** larger than within-group variability.

(b) State and interpret the p-value

p-value = **0.003**
Interpret the p-value in context. What does it mean, and what decision does it lead to?

✓ If H_0 were true (all group means equal), the probability of observing F ≥ 9.14
✓ by chance alone is 0.003. Because p-value = 0.003 < 0.05, we reject H_0.
✓ There is strong evidence that at least one study method is associated with a
✓ different mean exam score.

(c) Write a conclusion in context

Write a conclusion that addresses the research question. Use the phrase **"provides evidence"** rather than "proves." Do **not** claim causation unless the study design supports it.

✓ The data provide evidence that at least one study method is associated with a
✓ different mean exam score (F = 9.14, p = 0.003). Because students were randomly
✓ assigned to groups, this result supports a causal claim that study method
✓ affected exam performance. However, a post-hoc test would be needed to
✓ determine which specific methods differ.